

Numericals:

1. Two spherical raindrops of equal size are falling through air at a velocity of 0.08 m/s . If the drops join together forming a large spherical drop, what will be the new terminal velocity? (0.13 m/s)

Q₁

Chap 7

Data:-

$$V_T = 0.08 \text{ m/s}$$

New terminal velocity $V_{T'} = ?$

Sol:-

Before combination;

radius = r

Terminal velocity = V_T

After combination;

radius = R

Terminal velocity = $V_{T'}$

Since,

$$\frac{V_T}{V_{T'}} = \frac{r^2}{R^2} \rightarrow (i)$$

Volume before combination

= Volume after combination

$$2 \left(\frac{4}{3} \pi r^3 \right) = \frac{4}{3} \pi R^3$$

$$2r^3 = R^3$$

or

$$R^3 = 2r^3$$

$$R = (2r^3)^{\frac{1}{3}}$$

$$R = (2)^{\frac{1}{3}} r$$

put in eq (i)

$$\frac{0.08}{V'T} = \frac{r^2}{2^{\frac{1}{3}} \cdot r^2}$$

$$V'T = (2)^{\frac{2}{3}} \times (0.08)$$

$$V'T = 0.126 \text{ m/sec}$$

2. Calculate the viscous drag on a drop of oil of 0.1 mm radius falling through air at terminal velocity. (Viscosity of air = 1.8×10^{-5} Pa. s; density of oil = 850 kg/m^3)

($3.48 \times 10^{-8} \text{ N}$)

Q2

Data:-

$$r = 0.1 \text{ mm} = 0.1 \times 10^{-3}$$

$$\text{viscosity of air} = 1.8 \times 10^{-5} \text{ Pa}$$

$$\rho = \text{density of Oil} = 850 \text{ kg/m}^3$$

$$\text{viscous drag} = F_d = ?$$

Solution:-

At terminal velocity:-

$$F_{\text{air}} = \text{weight}$$

$$F_d = mg$$

$$F_d = \rho V g$$

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$$\therefore V = \frac{4}{3} \pi r^3$$

$$F_d = \rho \times \frac{4}{3} \times \pi \times r^3 \times g$$

$$F_d = 850 \times \frac{4}{3} (3.142) (0.1 \times 10^{-3})^3 \times 9.8$$

$$F_d = 3.48 \times 10^{-8} \text{ N}$$

3. What area must a heating duct have if air moving 3.0 m/s along it can replenish the air every 15 minutes in a room of volume 300 m³. Assume air density remains constant.
(0.11 m², 0.33 m)

Q 3

Data:-

$$V = 3 \text{ m/s}$$

$$t = 15 \text{ min} = 900 \text{ sec}$$

$$\text{Vol} = 300 \text{ m}^3$$

$$A = ?$$

Sol:-

Eq of Continuity:-

$$A_1 V_1 = A_2 V_2$$

$$AV = \frac{\Delta V}{\Delta T}$$

$$A = \frac{\Delta V}{V \cdot \Delta T}$$

$$A = \frac{300}{900 \times 3}$$

$$A = 0.11 \text{ m}^2$$

4. Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.50 m/s through a 4.0 cm diameter pipe in the basement under a pressure of 3.0 atm , what will be the flow speed and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.

(1.2 m/s , 2.5 atm)

Q4

$$V_1 = \text{speed} = 0.50 \text{ m/s}$$

$$d_1 = 4 \text{ cm} = 0.04 \text{ m}$$

$$P_1 = 3 \text{ atm} \times 1.01 \times 10^5 = 3.03 \times 10^5 \text{ N/m}^2$$

$$h_1 = 0$$

$$d_2 = 2.6 \text{ cm} = 0.026 \text{ m}$$

$$V_2 = ?$$

$$P_2 = ?$$

$$h_2 = 5 \text{ m}$$

Sol:-

$$r_1 = 0.04/2 = 0.02 \text{ m}$$

$$r_2 = 0.026/2 = 0.013 \text{ m}$$

$$\therefore A_1 V_1 = A_2 V_2$$

$$\pi r_1^2 V_1 = \pi r_2^2 V_2$$

$$V_2 = \left(\frac{r_1}{r_2} \right)^2 V_1$$

$$V_2 = \left(\frac{0.02}{0.013} \right)^2 (0.5)$$

$$V_2 = 1.183 \text{ m/s}$$

$$P_1 + \frac{1}{2} \rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g h_2$$

$$(3.03 \times 10^5) + \frac{1}{2} (1000) (0.5)^2 + (1000) (9.8) (0)$$

$$= P_2 + \frac{1}{2} (1000) (1.183)^2 + (1000) (9.8) (5)$$

$$3.03 \times 10^5 + 125 + 0 = P_2 + 1399.4 + 49000$$

$$303125 = P_2 + 50399.4$$

$$303125 - 50399.4 = P_2$$

$$P_2 = 253425.25 \text{ N/m}^2$$

or

$$P_2 = \frac{253425.25}{1.01 \times 10^5}$$

$$P_2 = 2.5 \text{ atm}$$

5. What is the volume rate of flow of water from a 1.85 cm diameter faucet if the pressure head is 12 m?
($4.6 \times 10^{-3} \text{ m}^3/\text{s}$)

Q5

Data:-

$$d = 1.85 \text{ cm} = 0.0185 \text{ m}$$

$$\frac{\Delta V}{\Delta T} = ?$$

$$F_1 = 12 \text{ m}$$

Sol:-

$$\text{Loss in P.E} = \text{gain in K.E}$$

$$mgh = \frac{1}{2}mv^2$$

$$2gh = v^2$$

$$v = \sqrt{2gh}$$

$$v = \sqrt{2(9.8)(12)}$$

$$v = 15.33 \text{ m/s}$$

Therefore,

$$\frac{\Delta V}{\Delta T} = A \times v$$

$$\frac{\Delta V}{\Delta T} = \pi r^2 \times v$$

$$\frac{\Delta V}{\Delta T} = (3.142) \left(\frac{d}{2}\right)^2 \times (15.33)$$

$$\frac{\Delta V}{\Delta T} = 4.1 \times 10^{-3} \text{ m}^3/\text{s}$$

6. The stream of water emerging from a faucet 'neck down' as it falls. The cross-sectional area is 1.2 cm^2 and 0.35 cm^2 . The two levels are separated by a vertical distance of 45 mm as shown in figure. At what rate does water flow from the tap? ($34 \text{ cm}^3 / \text{s}$)

Q6

$$A_1 = \frac{1.2 \text{ cm}^2}{10000} = 1.2 \times 10^{-4} \text{ m}^2$$

$$A_2 = 0.35 \text{ cm}^2 \times 10^{-4} = ~~0.35~~ 3.5 \times 10^{-5}$$

$$h = 45 \text{ mm} = 45 \times 10^{-3} \text{ m}$$

$$\frac{\Delta V}{\Delta T} = ?$$

Sol:-

$$\therefore A_1 V_1 = A_2 V_2$$

$$V_2 = \frac{A_1 V_1}{A_2}$$

$$V_2 = \frac{(1.2 \times 10^{-4}) V_1}{(0.35 \times 10^{-4})}$$

$$V_2 = 3.428 V_1 \rightarrow (i)$$

Loss in P.E = Gain in P.E

$$mgh = \frac{1}{2} m V_2^2 - \frac{1}{2} m V_1^2$$

$$gh = \frac{1}{2} (V_2^2 - V_1^2)$$

$$(9.8)(45 \times 10^{-3}) = \frac{1}{2} [(3.428 V_1)^2 - V_1^2]$$

$$2(0.441) = 11.75 V_1^2 - V_1^2$$

$$0.882 = 10.696 V_1^2$$

$$V_1^2 = \frac{0.882}{10.696}$$

$$V_1 = \sqrt{0.082}$$

$$V_1 = 0.287 \text{ m/sec}$$

$$\therefore \frac{\Delta V}{\Delta T} = A_1 \times V_1$$

$$\frac{\Delta V}{\Delta T} = (1.2 \times 10^{-4})(0.287)$$

$$\frac{\Delta V}{\Delta T} = 3.4 \times 10^{-5} \text{ m}^3/\text{s}$$

or,

$$\frac{\Delta V}{\Delta T} = 3.4 \times 10^{-5} \times 10^6$$

$$\frac{\Delta V}{\Delta T} = 34.4 \text{ cm}^3/\text{s}$$

7. Water leaves the jet of a horizontal hose at 10 m/s. If the velocity of water within the hose is 0.40 m/s, calculate the pressure within the hose. Density of water is 1000 kg/m^3 and atmospheric pressure is 100000 Pa . ($1.5 \times 10^5 \text{ Pa}$)

Q7

$$V_1 = 10 \text{ m/s}$$

$$V_2 = 0.4 \text{ m/s}$$

$$\rho_{\text{water}} = 1000 \text{ kg/m}^3$$

nc2

$$P_1 = 100000 = 10^5 \text{ Pa}$$

$$P_2 = ?$$

Sol: $P_1 + \frac{1}{2} \rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g h_2$

Since hose is horizontal then $h = 0$

$$P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2$$

$$10^5 + \frac{1}{2} (1000)(10)^2 = P_2 + \frac{1}{2} (1000)(0.4)^2$$

$$150000 = P_2 + 80$$

$$P_2 = 150000 - 80$$

$$P_2 = 1.499 \times 10^5 \text{ Pa}$$

8. What is the maximum weight of an aircraft with a wing area of 50 m^2 flying horizontally if the velocity of the air over the upper surface of the wing is 150 m/s and that the lower surface 140 m/s ? Density of air is 1.29 kg/m^3 ($9.3 \times 10^4 \text{ N}$)

Q8 Data:-

$$W = ?$$

$$A_1 = 50 \text{ m}^2$$

$$V_1 = 150 \text{ m/s} \quad (\text{upper surface of wing})$$

$$V_2 = 140 \text{ m/s} \quad (\text{lower surface of wing})$$

$$\rho = 1.29 \text{ kg/m}^3$$

Sol:-

$$\therefore P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

moving is horizontal so $h = 0$

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$(P_1) + \frac{1}{2} (1.29) (150)^2 = P_2 + \frac{1}{2} (1.29) (40^2)$$

$$P_1 + 14512.5 = P_2 + 12642$$

$$P_2 - P_1 = 14512.5 - 12642$$

$$\Delta P = 1870.5$$

$$\text{upward force} = F_2 - F_1$$

$$= P_2 A - P_1 A$$

$$= A (P_2 - P_1)$$

$$= A \Delta P$$

$$= (1870.5) (50)$$

$$= 93525$$

or

$$9.3 \times 10^4 \text{ N}$$

Upward force is w of Aircraft = $9.3 \times 10^4 \text{ N}$

9. A liquid flows through a pipe with a diameter of 0.50 m at a speed of 4.20 m/s. What is the rate of flow in L/min? (49500 L/min)

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Q9

$$d = 0.5 \text{ m}$$

$$V = 4.2 \text{ m/s}$$



$$\text{rate flow in L/min} = \frac{\Delta V}{\Delta T} = ?$$

$$\Rightarrow \frac{\Delta V}{\Delta T} = A \times V$$

$$\Rightarrow \frac{\Delta V}{\Delta T} = \pi \left(\frac{d}{2} \right)^2 \times (4.2)$$

$$\Rightarrow \frac{\Delta V}{\Delta T} = 0.8246 \text{ m}^3/\text{sec}$$

$$1000 \text{ Litres/s} = 1 \text{ meter cube/s}$$

$$0.8246 \times 1000 \text{ L/s} = 1 \times 0.8246 \text{ m}^3/\text{s}$$

$$824.6 \text{ L/s} = 0.8246 \text{ m}^3/\text{s}$$

$$\frac{\Delta V}{\Delta T} = \frac{824.6}{\frac{1}{60}}$$

$$\frac{\Delta V}{\Delta T} = 49476 \text{ L/s}$$

$$L = 5.4 \times 10^5 \text{ N}$$

Chap 7

Q10 Calculate the average speed of blood flow in the major arteries of the body, which have a ~~total cross~~ radius sectional of about 2.1 cm^2 . Use data of example.

Data:-

$$\text{radius of artery} = r_1 = 2.1 \text{ cm} = 0.021 \text{ m}$$

Speed of blood = ?

$$\text{radius of capillary} = r_2 = 4 \times 10^{-4} \text{ cm} = 4 \times 10^{-6} \text{ m}$$

$$\text{speed in capillary} = V_2 = 5 \times 10^{-4} \text{ m/s}$$

$$\text{No. of capillaries} = N = 7 \times 10^9$$

Sol:-

A/c to equation of continuity

$$A_1 V_1 = N A_2 V_2$$

$$\pi r_1^2 V_1 = N \pi r_2^2 V_2$$

$$V_1 = \frac{N r_2^2 V_2}{r_1^2}$$

$$V_1 = \frac{(7 \times 10^9)(4 \times 10^{-6})^2(5 \times 10^{-4})}{(0.021)^2}$$

$$V_1 = 0.126 \text{ m/s} \times 100$$

$$V_1 = 12.6 \text{ cm/s}$$

If you find any mistake in these, Kindly report it to the Officials of NCA